ECE 340 # Homework Assignment #5 **Fall 2026**

DUE: FRIDAY, OCTOBER 2, 2026

MAKE SURE YOUR **NAME** AND **NETID** ARE CLEARLY VISIBLE. SUBMIT ASSIGNMENTS AS A PDF (EITHER SCANNED OR TYPED). SHOW ALL WORK (PARTIAL CREDIT CANNOT BE GIVEN WITHOUT WORK). SHOW ALL UNITS. HOMEWORK MUST BE TURNED IN THROUGH GRADESCOPE. HOMEWORK SHOULD BE TURNED IN ON TIME BEFORE 5:00PM ON THE DUE DATE.

1. STEADY STATE CARRIER GENERATION; QUASI-FERMI LEVEL

A Si sample is doped with $N_D = 2 * 10^{16}$ cm^{-3} and optically excited uniformly with $g_{op} = 10^{19}$ EHP/cm^3·s. Assume T = 300 K and use thermal generation rate $g(T) = 4.5 \times 10^{10} cm^{-3}$/s and diffusion coefficients $D_n = 30$ cm^2/s and $D_p = 10$ cm^2/s.

(A). Find the thermal equilibrium electron and hole concentrations n_0 and p_0.

(B). State the low-level injection and steady-state band-to-band recombination assumptions. Using these assumptions, simplify the generation-recombination balance equation $g(T) + g_{op} = \alpha_r np$ into an expression for the excess carrier concentration δp.

(C). Solve for δp.

(D). Determine the steady-state electron hole concentrations n and p. Check that the low-level injection assumption from (B) is consistent with your answer.

(E). Calculate the separation of the electron and hole quasi-Fermi levels $(F_n - F_p)$.

(F). Calculate the change in conductivity of the Si upon illumination.

Solution **20 points total**

(A). **2 points total.**

$$n_0 \approx N_D = 2 \times 10^{16} cm^{-3}$$
$$p_0 = \frac{n_i^2}{n_0} = \frac{(1.5 \times 10^{10})^2}{2 \times 10^{16}} = 1.125 \times 10^4 cm^{-3}$$

(B). **6 points total.**

Steady-state band-to-band recombination: the excess electrons and holes are created and recombine in pairs, so $\delta n = \delta p$, and recombination is direct, so the recombination rate is $\alpha_r np$.

Low-level injection: the excess carrier density is assumed small compared to the majority carrier concentration ($\delta p << n_0$), so the majority carrier concentration is essentially unperturbed ($n \approx n_0$).

Applying both assumptions to the balance equation:

$$\begin{aligned} g(T) + g_{op} &= \alpha_r np = \alpha_r (n_0 + \delta n)(p_0 + \delta p) \\ &\approx \alpha_r n_0 p_0 + \alpha_r n_0 \delta p \end{aligned}$$

Since $g(T) = \alpha_r n_0 p_0$ by the definition of thermal equilibrium

$$\begin{aligned} g_{op} &= \alpha_r n_0 \delta p \approx \alpha_r (n_0 + p_0)\delta p \\ \delta p &\approx \frac{g_{op}}{\alpha_r (n_0 + p_0)} \end{aligned}$$

(C). **3 points total.**

Using $g(T) = \alpha_r n_i^2$ and the expression found in part (B).

$$\delta p = \frac{g_{op} n_i^2}{g(T)(n_0 + p_0)} = \frac{(1 \times 10^{19})(1.5 \times 10^{10})^2}{(4.5 \times 10^{10})(2 \times 10^{16})} = 2.5 \times 10^{12} cm^{-3}$$

(D). **3 points total.**

$$\begin{aligned} n &= n_0 + \delta n = 2 \times 10^{16} + 2.5 \times 10^{12} \approx 2 \times 10^{16} cm^{-3} \\ p &= p_0 + \delta p = 1.125 \times 10^{4} + 2.5 \times 10^{12} \approx 2.5 \times 10^{12} cm^{-3} \end{aligned}$$

The low-level injection assumption is satisfied as $\delta p << n_0$. δp is roughly four orders of magnitude smaller than n_0.

(E). **3 points total.**

$$\begin{aligned} F_n - E_i &= kTln(\frac{n}{n_i}) = 0.0259 ln(\frac{2 \times 10^{16}}{1.5 \times 10^{10}}) = 0.365 eV \\ E_i - F_p &= kTln(\frac{p}{n_i}) = 0.0259 ln(\frac{2.5 \times 10^{12}}{1.5 \times 10^{10}}) = 0.133 eV \\ \implies F_n - F_p &= 0.365 + 0.133 = 0.498 eV \end{aligned}$$

(F). **3 points total.**

Calculating mobilities using the Einstein relation

$$\mu_n = \frac{D_n q}{kT} = \frac{30}{0.0259} \approx 1158 cm^2/V \cdot s$$
$$\mu_p = \frac{D_p q}{kT} = \frac{10}{0.0259} \approx 386 cm^2/V \cdot s$$

The change in conductivity can be computed using the excess carrier concentrations.

$$\Delta\sigma = q(\delta n \mu_n + \delta p \mu_p) = (1.6 \times 10^{-19})(2.5 \times 10^{12})(1158 + 386) \approx 6.18 \times 10^{-4} S \cdot cm^{-1}$$

2. Diffusion and Drift of Carriers

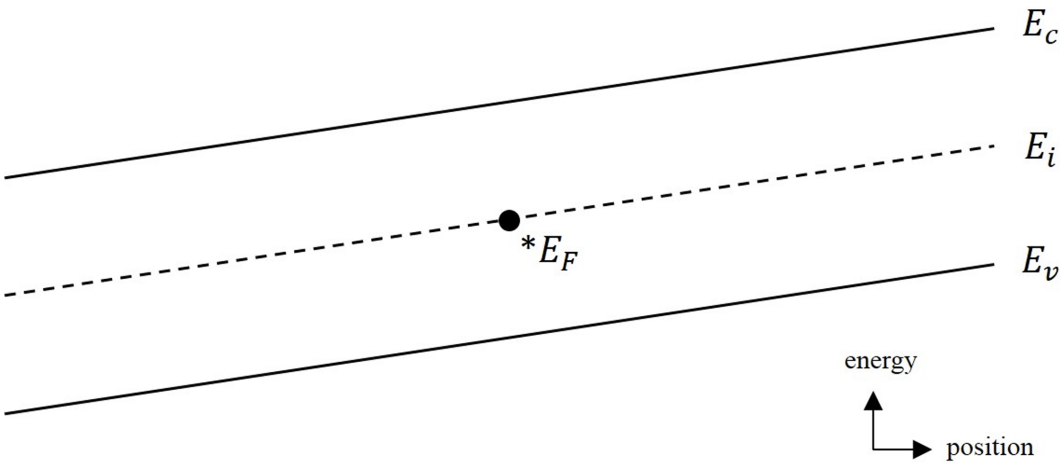


Figure 2.1: Band diagram under electric field bias

Consider the **thermal equilibrium** band diagram of Figure 2.1 for a portion of a semiconductor sample with a built-in electric field $\boldsymbol{E}$.

(A). Sketch the Fermi level as a function of position *through* the indicated point, E_F, across the width of the band diagram of Figure 2.1.

(B). Based on Figure 2.1, sketch and label both logarithmic electron and hole concentrations as a function of position across the full width of the sample. Label the slopes for each and the location of $\log(n_i)$. *Note:* the carrier concentration scale should be $\log(n), \log(p)$ on the same graph.

(C). Determine the direction of the electric field $\boldsymbol{E}$. Indicate the direction of **electron flows** and the corresponding **drift current** and **diffusion current** directions caused by electrons.

(D). Repeat (C) for **holes**.

Solution **20 points in total**

(A). **2 points in total.**

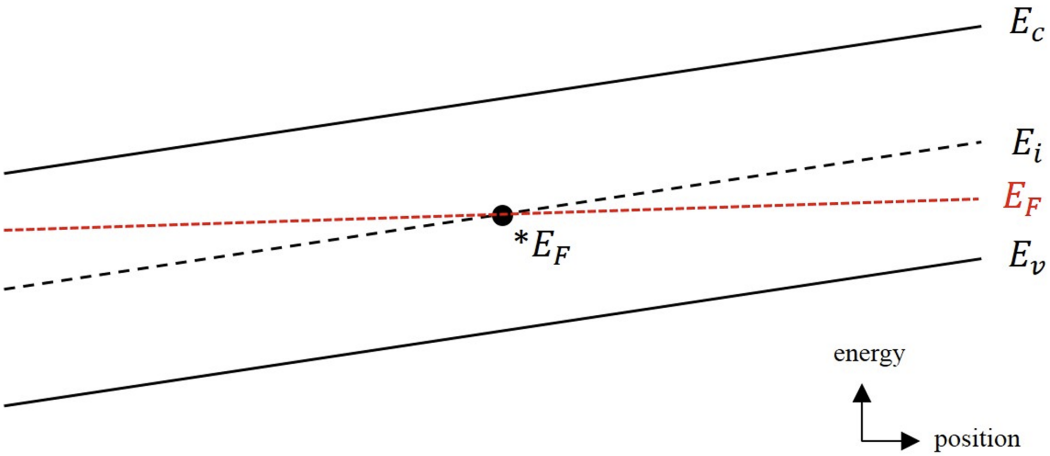


Figure 2.2: Thermal eqilibrium, so flat Fermi level

(B). **10 points in total.**

From the equations

$$n_0 = n_i e^{(E_F - E_i)/kT}$$
$$p_0 = n_i e^{(E_i - E_F)/kT}$$

one can get the slopes

$$\log(n_0) = \log(n_i) + \frac{E_F - E_i}{kT}$$
$$\frac{d(\log(n_0))}{dx} = \frac{d(\log(n_i))}{dx} + \frac{d}{dx}\left(\frac{E_F - E_i}{kT}\right) = \frac{d}{dx}\left(\frac{-E_i}{kT}\right)$$

and

$$\log(p_0) = \log(n_i) + \frac{E_i - E_F}{kT}$$
$$\frac{d(\log(p_0))}{dx} = \frac{d(\log(n_i))}{dx} + \frac{d}{dx}\left(\frac{E_i - E_F}{kT}\right) = \frac{d}{dx}\left(\frac{E_i}{kT}\right)$$

At the indicated point where $E_F = E_i$, the values $E_F - E_i = 0$ and $E_i - E_F = 0$, showing that the carrier concentrations must be equal to the intrinsic value at that position.

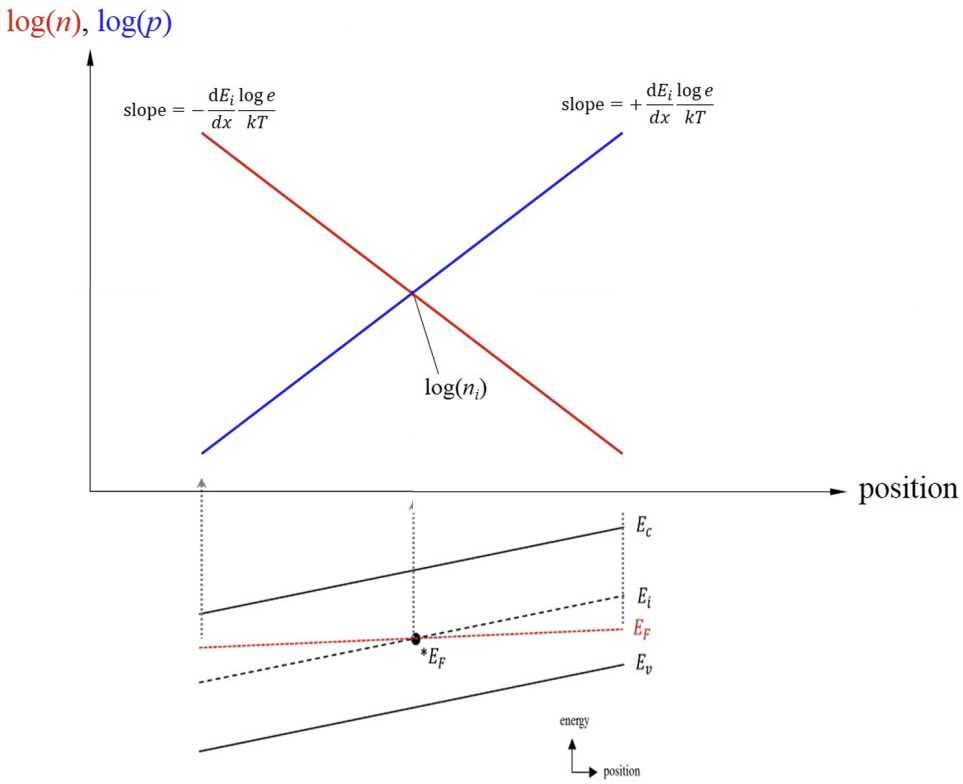


Figure 2.3: Solution showing carrier concentrations as a function of position.

(C). AND (D). **8 points in total.**

The electric field is in the +position direction, as $\boldsymbol{E}(x) = -\frac{dV(x)}{dx} = \frac{1}{q}\frac{dE_i}{dx}$.

	Drift Flow	Diffusion Flow	J_{drift}	$J_{diffusion}$
Electrons	⇐	⇒	⇒	⇐
Holes	⇒	⇐	⇒	⇐

Table 2.1: Electron and hole flow and current directions

3. ELECTRON DIFFUSION IN INP

At 300 K, a piece of InP has a majority electron concentration profile

$$n(x) = N_D e^{-x/\alpha}$$

where $\alpha > 0$.

(A). With a cross-sectional Area A_0 in the y-z plane, derive an expression for the electron diffusion current flowing through the piece.

(B). What is the electron's diffusion direction? Explain.

(C). Assume the sample is at equilibrium; calculate an expression for the electric field.

(D). The electron mobility and lifetime of InP at 300 K are 4000 $\text{cm}^2/(V \cdot s)$ and $\tau_n = 8\ \mu$s, respectively. Determine the diffusion constant D_n and the diffusion length L_n.

(E). The diffusion equation for electrons is

$$\frac{\partial \delta n}{\partial t} = D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\partial n}{\tau_n},$$

and the diffusion current density is

$$J_n = qD_n \frac{\partial \delta n}{\partial x}.$$

Give the definition of *steady state* and apply it to the diffusion equation to reach a simplified version which includes diffusion length.

(F). Write the general form for the solution of the partial differential equation derived in (E) as a sum of two exponentials. *Hint: see Sec. 4.4.4 in the book.*

(G). Apply the following boundary conditions to the solution derived in (F):

$$\delta n(0) = \Delta n, \qquad \delta n(L) = 0,$$

where L is the sample size. You do not need to solve these two equations; applying the boundary conditions will be enough.

Solution **20 points in total**

(A). **2 points in total.**

$$J_{n,diff} = qD_n \frac{dn(x)}{dx} = -\frac{qD_n N_D}{\alpha} e^{-x/\alpha}$$
$$I_{n,diff} = A_0 \times J_{n,diff} = -\frac{A_0 qD_n N_D}{\alpha} e^{-x/\alpha}$$

(B). **4 points in total.**

The electron diffusion current is in the $-x$ direction, as derived in part (A). Current is defined as the direction of positive charge movement, and as electrons are negatively charged, they must diffuse in the $+x$ direction.

Alternative explanation: Net diffusion occurs from regions of high particle concentration to regions of low particle concentration. Since $\alpha > 0$, electron concentration decreases with x, so electrons diffuse in the $+x$ direction.

(C). **4 points in total.**

In equilibrium, the net current (diffusion + drift) must be zero:

$$J_n(x) = 0 = J_{n,diff} + J_{n,drift} = qD_n \frac{dn(x)}{dx} + q\mu_n n(x)E(x)$$
$$\Rightarrow -qD_n \frac{dn(x)}{dx} = q\mu_n n(x)E(x)$$

Thus,

$$E(x) = -\frac{D_n}{\mu_n}\frac{1}{n(x)}\frac{dn(x)}{dx} = \frac{D_n}{\mu_n \alpha} = \frac{kT}{q\alpha}$$

(D). **4 points in total.**

$$D_n = \frac{kT}{q}\mu_n = 0.0259 \times 4000 = 103.6 \text{ cm}^2/\text{s}$$
$$L_n = \sqrt{\tau_n D_n} = \sqrt{(8 \times 10^{-6})(103.6)} \approx 2.88 \times 10^{-2} \text{ cm} = 288\mu\text{m}$$

(E). **2 points in total.**

Steady state means no time dependence, so

$$\frac{\partial^2 \delta n}{\partial x^2} = \frac{\partial n}{\tau_n D_n} = \frac{\partial n}{L_n^2}$$

(F). **2 points in total.**

The solution to this second-order ODE is exponential:

$$\delta n(x) = Ae^{x/L_n} + Be^{-x/L_n}$$

(G). **2 points in total.**

Apply the boundary conditions:

$$\delta n(0) = A + B = \Delta n$$
$$\delta n(L) = Ae^{L/L_n} + Be^{-L/L_n} = 0$$

4. Carrier Diffusion in p-type Silicon

In a very long p-type Si sample extending along the x-axis with cross-sectional area $A = 1$ cm^2, electrons are injected such that the steady-state excess electron concentration is 1×10^{18} cm^{-3} at $x = 0$.

(A). Given $\mu_n = 1000$ cm$^2/(V \cdot s)$ and $\tau_n = 100$ ps, calculate the diffusion coefficient D_n and the diffusion length L_n of this sample.

(B). Write an expression for the excess electron concentration Δn with respect to position x.

(C). Using integration, calculate the total excess electron charge (storage charge) Q_n. Then, provide an intuitive interpretation of diffusion length.

(D). Write an expression for the electron diffusion current I_n as a function of position x. What is the maximum electron diffusion current and at what position does it occur? What is the electron diffusion current at $x = \infty$?

(E). From common sense, current in a metal wire is the same at any position (current continuity). Does your answer to part (D) violate current continuity? Why or why not? *Hint: think about the meaning of* τ_n.

Solution **20 points in total**

(A). **4 points in total.**

$$D_n = \frac{kT}{q}\mu_n = 0.0259 \times 1000 = 25.9 \text{ cm}^2/\text{s}$$
$$L_n = \sqrt{\tau_n D_n} = \sqrt{25.9 * 10^{-10}} = 5.089 \times 10^{-5} \text{ cm} = 0.509 \mu\text{m}$$

(B). **2 points in total.**

$$\delta n(x) = \delta n(0) e^{-x/L_n}$$

(C). **6 points in total.**

$$Q_n = qA\int_0^\infty \delta n(x)dx = qAL_n\delta n(0)$$
$$= 1.6 \times 10^{-19} * 1 * 5.089 \times 10^{-5} * 10^{18} = 8.14 \times 10^{-6} \text{ C}$$

Interpretation: The integration effectively multiplies the starting excess carrier density $\delta n(0)$ by the diffusion length. Thus, L_n represents the average distance a carrier travels before recombining.

(D). **4 points in total.**

$$I_n(x) = -qAD_n\frac{\partial \delta n}{\partial x} = qA\frac{D_n}{L_n}\delta n(0)e^{-x/L_n}$$

The maximum diffusion current occurs at $x = 0$:

$$I_n(0) = qA\frac{D_n}{L_n}\delta n(0) = 1.6\times10^{-19} * 1 * \frac{25.9}{5.089\times10^{-5}} * 10^{18} = 8.14\times10^4 \text{ A}$$

At $x = \infty$, the diffusion current vanishes:

$$I_n(\infty) = 0$$

(E). **4 points in total.**

No, current continuity is not violated. In part (D), current decreases with x because electrons are constantly recombining with holes (characterized by lifetime τ_n). Thus, carriers are lost to recombination, and the diffusion current naturally diminishes with distance. This is consistent with continuity when recombination is included: carriers are not disappearing arbitrarily, but recombining.